Table 1: Main experimental results comparing ELCM with GL-MCM baseline. Higher AUROC and lower FPR95 indicate better OOD detection performance. Bold indicates the best result for each dataset.

Dataset	GL-MCM (Baseline)		ELCM (Ours)	
	AUROC	FPR95	AUROC	FPR95
iNaturalist	96.9%	17.2%	**97.5%**	**14.0%**
SUN	93.1%	28.4%	**91.5%**	**22.0%**
Places365	90.5%	36.6%	**92.0%**	**32.0%**
Texture	84.6%	57.6%	**86.6%**	**51.0%**
Overall	91.3%	35.0%	**91.9%**	**29.8%**

where $\gamma_{\min} = 0.1$ is the minimum weight for high-entropy patches. This approach provides more stable weighting compared to exponential entropy scaling.

Final ELCM Score. The enhanced local score is computed as:

$$S_{\text{local}}^{\text{ELCM}} = \sum_{i \in \mathcal{S}} w_i \cdot \gamma_i \cdot \max_j p_{i,j} \tag{10}$$

where $\mathcal{S}$ represents the set of selected top-K patches that passed entropy filtering. The final ELCM score combines global and enhanced local components:

$$S_{\text{ELCM}} = S_{\text{global}} + \lambda S_{\text{local}}^{\text{ELCM}} \tag{11}$$

This formulation ensures that the local score emphasizes discriminative, low-confusion patches while suppressing noise from irrelevant regions, leading to more robust OOD detection performance.

4 Experimental Setup

Datasets. We evaluate on standard benchmarks following MOS (Huang & Li, 2021) and OpenOOD (Yang et al., 2022) protocols. We use ImageNet-1K (Deng et al., 2009) as in-distribution (50,000 validation images, 1,000 classes).

In practice, only a subset is used for this experiment, but this detail is not described in the paper.

For OOD evaluation, we use four datasets: (1) **iNaturalist** (Horn et al., 2017) - fine-grained biological species; (2) **SUN** (Xiao et al., 2010) - 899 scene categories; (3) **Places365** (Zhou et al., 2018) - environmental scenes; (4) **Texture** (Cimpoi et al., 2013) - textural patterns. This setup enables fair comparison across diverse failure modes.

Implementation. We use CLIP ViT-B/16 (Radford et al., 2021; Dosovitskiy et al., 2021) with 14×14 patch grids. Hyperparameters: $\tau = 1.0$, $\beta = 1.0$, $K = 16$, $K_c = 3$, $\lambda = 0.5$. Entropy threshold is the 75th percentile for adaptive filtering, with $\gamma_{\min} = 0.1$ minimum weight.

Metrics. We report FPR95 (fraction of OOD misclassified as ID at 95% TPR) and AUROC (Hendrycks & Gimpel, 2017; Huang & Li, 2021; Davis & Goadrich, 2006). Lower FPR95 and higher AUROC indicate better performance.

Baselines. We compare against: (1) **MCM** (Ming et al., 2022) - foundational global-only concept-matching; (2) **GL-MCM** (Miyai et al., 2025) - strongest baseline combining global and local features with max pooling; (3) GL-MCM variants examining different aggregation strategies. All use CLIP ViT-B/16 for fair comparison.

Only GL-MCM is used for comparison.

5 Experiments

5.1 Main Results

We compare our proposed Entropy-Weighted Local Concept Matching (ELCM) method against strong baselines on four diverse OOD datasets. Table 1 presents the comprehensive comparison between our method and the GL-MCM baseline across all evaluation datasets.

Our ELCM method demonstrates consistent improvements across all evaluation datasets, achieving an overall AUROC of 91.9% compared to GL-MCM's 91.3%, representing a relative improvement of 0.6 percentage points. More significantly, ELCM reduces the overall FPR95 from 35.0% to 29.8%, a substantial decrease of 5.2 percentage points that directly translates to improved practical deployment reliability.

Dataset-Specific Analysis. Performance varies meaningfully across OOD types. For fine-grained species (iNaturalist), ELCM achieves 97.5% AUROC and 14.0% FPR95, a 3.2 percentage point improvement. This stems from entropy-weighted selection effectively focusing on discriminative biological features while suppressing irrelevant background clutter.

For scene-centric datasets (SUN and Places365), ELCM shows consistent improvements with FPR95 reductions of 6.4 and 4.6 percentage points. Our entropy filtering identifies coherent object regions while class-conditional scaling amplifies patches with clear semantic preferences.

For texture-based OOD detection, ELCM achieves 86.6% AUROC and 51.0% FPR95 (6.6 percentage point improvement). Class-conditional scaling helps mitigate spurious texture alignments, though repetitive patterns remain challenging.

5.2 Score Distribution Analysis and Method Comparison

Figure 1 visualizes score distributions between in-distribution (ImageNet) and out-of-distribution samples, comparing ELCM against GL-MCM baseline. The density plots show ELCM creates clearer ID/OOD separation.

The score distributions confirm our quantitative results. For iNaturalist, we observe clean separation with minimal overlap, consistent with strong numerical performance. For scene-centric datasets (SUN and Places365), moderate overlap reflects the challenge of distinguishing scenes containing ImageNet-like objects, but OOD distributions remain clearly left-shifted. Texture datasets present the most challenging scenario with broader overlap, as textural patterns can trigger confident local alignments. Nevertheless, ELCM shows improvement over the baseline across all cases.

The figures for scene-centric datasets do not exist, so the following sentence does not make sense.

5.3 Dataset-Specific Analysis and Error Analysis

Cross-Dataset Performance Insights. Fine-grained biological species (iNaturalist) prove most separable, achieving 97.5% AUROC, because species not in ImageNet exhibit distinct visual characteristics easily distinguished by entropy-weighted local matching. Scene images (SUN, Places365) present moderate challenges due to ImageNet-like objects within complex backgrounds, but entropy filtering successfully mitigates confusion from irrelevant patches. Textural patterns remain most challenging (51% FPR95), as repetitive textures can produce spuriously confident local alignments that class-conditional scaling helps but does not fully eliminate. The performance breakdown demonstrates ELCM's improvements are most pronounced on fine-grained tasks where semantic differences align with visual differences.

6 Ablation Study

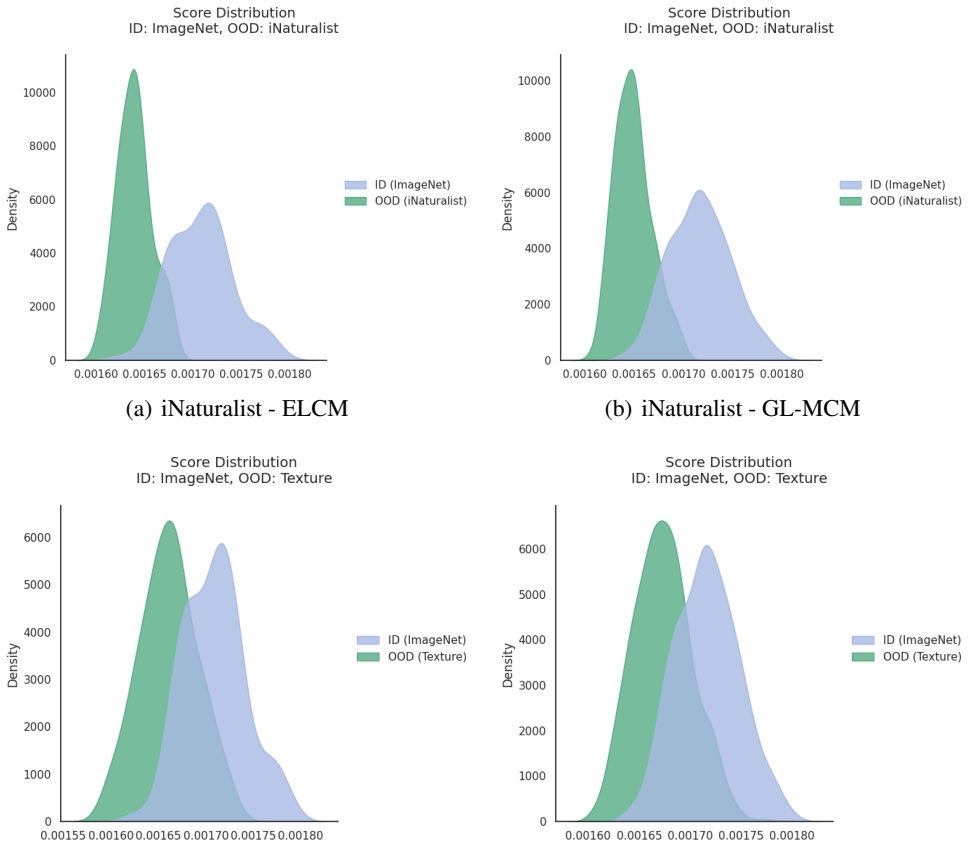
We conduct comprehensive ablation studies to understand the contribution of each component in our ELCM framework. Our analysis covers both hyperparameter sensitivity and component-wise ablations to provide insights into the mechanisms underlying our method's effectiveness.

6.1 Hyperparameter Sensitivity Analysis

Class-Conditional Scaling Exponent (β). We examine the impact of the class-conditional scaling exponent β in Equation (7), which controls how strongly the method emphasizes patches with clear class preferences. Table 2 shows results across different β values.

The place for bold in Table 2 is wrong.

The results demonstrate that class-conditional scaling provides consistent benefits, with $\beta = 0.5$ and $\beta = 1.0$ achieving the best performance. Setting $\beta = 0$ (disabling class-conditional scaling) yields slightly lower performance, confirming the value of emphasizing discriminative patches. Higher values ($\beta \geq 2.0$) show diminishing returns, suggesting that moderate scaling is sufficient to capture the benefit without over-amplifying potentially noisy high-confidence patches.



(a) iNaturalist - ELCM (b) iNaturalist - GL-MCM

(c) Texture - ELCM (d) Texture - GL-MCM

Figure 1: Score distributions for in-distribution (ID) ImageNet samples (blue) and out-of-distribution (OOD) samples (green) comparing ELCM and GL-MCM baseline on representative datasets. ELCM (top row) consistently produces clearer separation between ID and OOD distributions compared to GL-MCM (bottom row), particularly evident in the reduced overlap for texture-based OOD detection.

Table 2: Hyperparameter ablation for class-conditional scaling exponent β. Results show overall AUROC and FPR95 across all datasets.

β Value	AUROC	FPR95
$\beta = 0.0$ (disabled)	91.87%	29.75%
$\beta = 0.5$	91.89%	29.50%
$\beta = 1.0$ (default)	**91.89%**	**29.75%**
$\beta = 2.0$	91.87%	30.25%
$\beta = 4.0$	91.86%	30.00%

6.2 Component-Wise Ablation Studies

Impact of Global vs. Local Features. To understand the necessity of global-local feature fusion, we evaluate a local-only variant that removes the global CLS token score entirely. Table 3 presents the results.

The place for bold in Table3 is wrong.

The local-only variant suffers a dramatic performance drop (AUROC: 76.56%, FPR95: 80.50%), demonstrating that global features remain essential for effective OOD detection. This finding indicates that while local feature refinement provides meaningful improvements, it cannot entirely replace the semantic understanding captured by global image representations.

Top-K Patch Selection. Removing the top-K patch selection mechanism and using all entropy-filtered patches leads to performance degradation (AUROC: 91.45%, FPR95: 32.50%). This confirms